CSE113
Foundations of Computer Science

State University of New York, Korea

Agenda

- Size of infinite sets

- Five classic examples

Cardinality/Size of Infinite sets

- There are as many squares as there are numbers because they are just as numerous as their roots.

- — Galileo Galilei, 1632

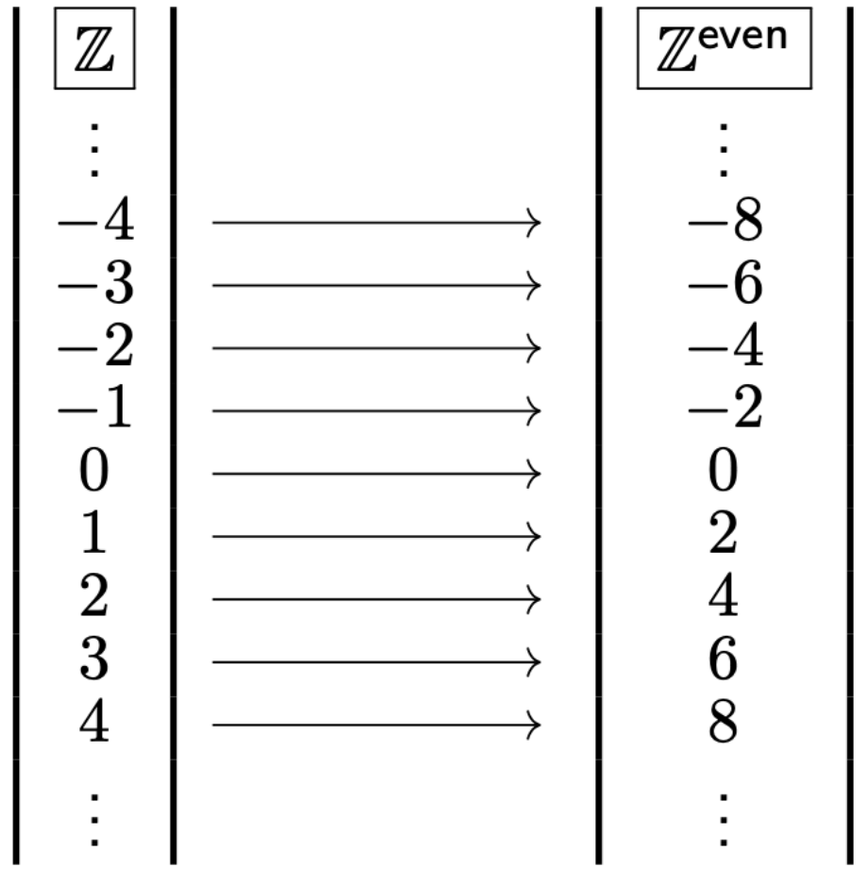
Same cardinality

Definition

- Let A and B be any sets. A has the same cardinality as B if, and only if, there is a one-to-one correspondence from A to B.
- A has the same cardinality as B if, and only if, there is a function f from A to B that is both one-to-one and onto.

Example 1

Integers and even numbers are of the same size



Proof

Problem

- Prove that the cardinality of integers and even numbers are the same.

Solution

Problem

- Prove that the cardinality of integers and even numbers are the same.

- To prove that $|\mathbb{Z}| = |\mathbb{Z}^{\text{even}}|$, we need to prove that there is a one-to-one correspondence, say f, between $\mathbb{Z}$ and $\mathbb{Z}^{\text{even}}$. Suppose $f = 2n$ for all integers $n \in \mathbb{Z}$.
- Prove that f is one-to-one.
  Suppose $f(n_1) = f(n_2)$.
  $\implies 2n_1 = 2n_2$ ($\because$ Defn. of f)
  $\implies n_1 = n_2$ ($\because$ Simplify)
- Prove that f is onto.
  Suppose $m \in \mathbb{Z}^{\text{even}}$.
  $\implies m$ is even ($\because$ Defn. of $\mathbb{Z}^{\text{even}}$)
  $\implies m = 2k$ for $k \in \mathbb{Z}$ ($\because$ Defn. of even)
  $\implies f(k) = m$ ($\because$ Defn. of f)

An infinite set and its proper subset can have the same size!



Countable sets

$\mathbb{N}$		A
1	$\longrightarrow$	"First" element of A
2	$\longrightarrow$	"Second" element of A
3	$\longrightarrow$	"Third" element of A
4	$\longrightarrow$	"Fourth" element of A
5	$\longrightarrow$	"Fifth" element of A
.		.
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Definition

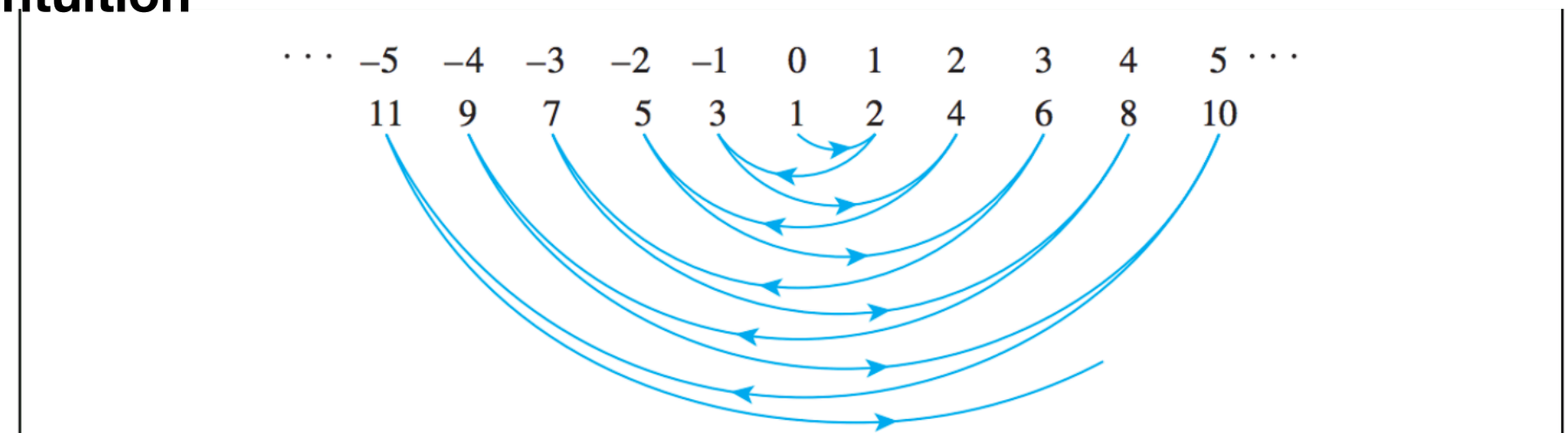
- A set is called countably infinite if, and only if, it has the same cardinality as the set of positive integers.
- A set is called countable if, and only if, it is finite or countably infinite. A set that is not countable is called uncountable.

Example 2

Problem

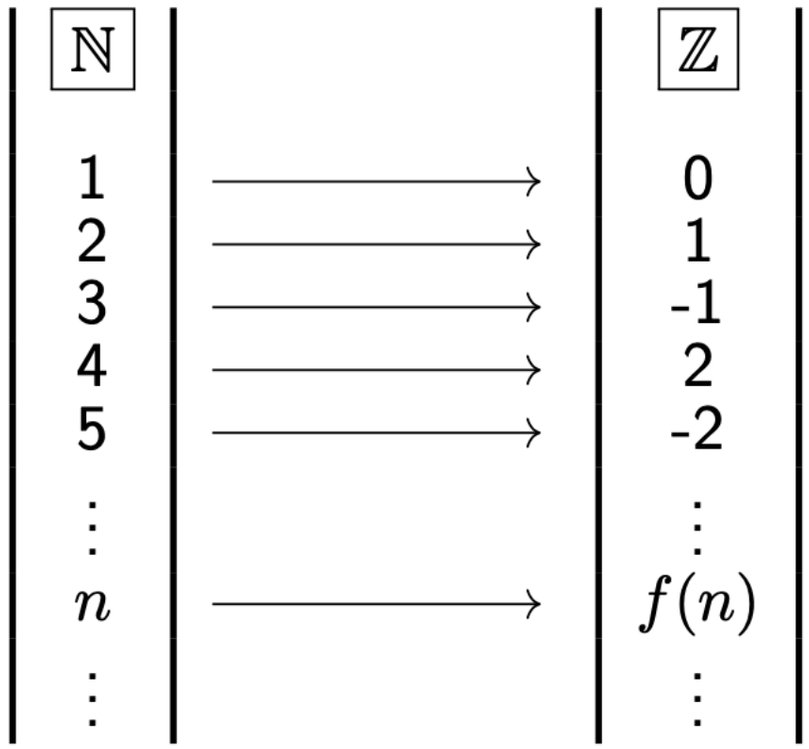
- Prove that the set of integers is countably infinite.

Intuition



Integers are countable

Solution (continued)



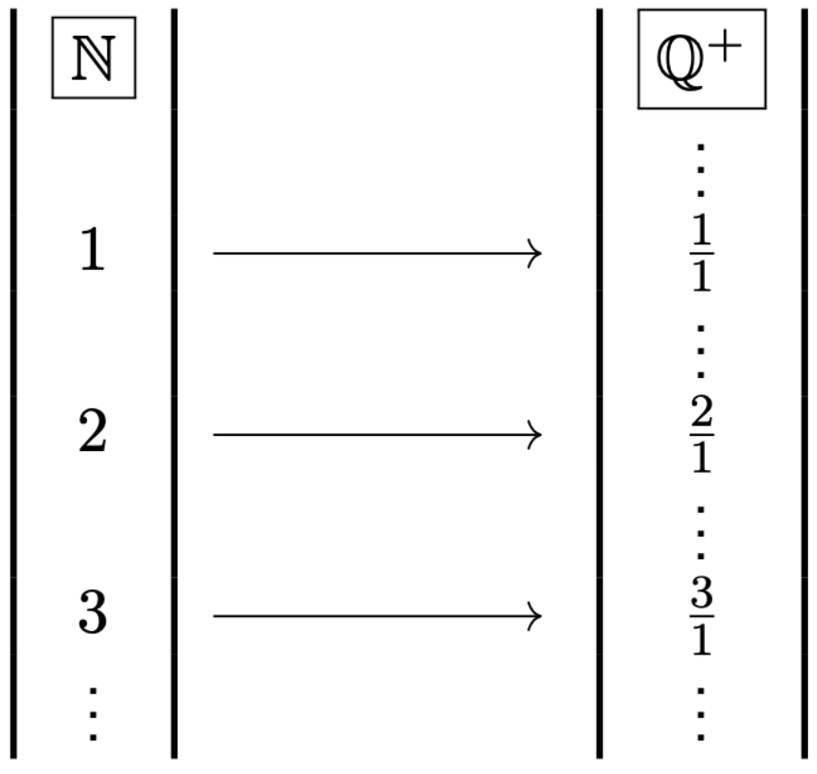
- Define a function $f(n) : \mathbb{N} \to \mathbb{Z}$ such that

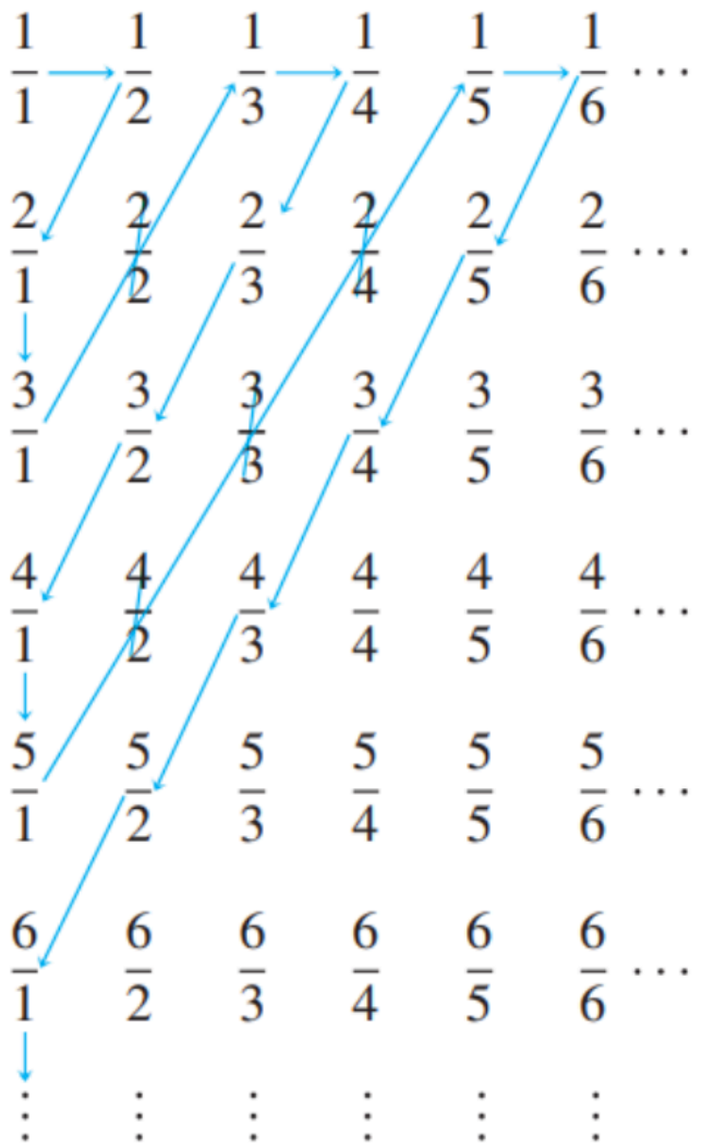
$$f(n) = \begin{cases} \frac{n}{2} & \text{if } n \text{ is an even natural number,} \\ -\left(\frac{n-1}{2}\right) & \text{if } n \text{ is an odd natural number.} \end{cases}$$

- As f is a one-to-one correspondence between $\mathbb{N}$ and $\mathbb{Z}$, the set of integers is countably infinite.

Example 3: True or False?

Set of positive rationals is uncountable





Set of positive rationals is countable

Problem

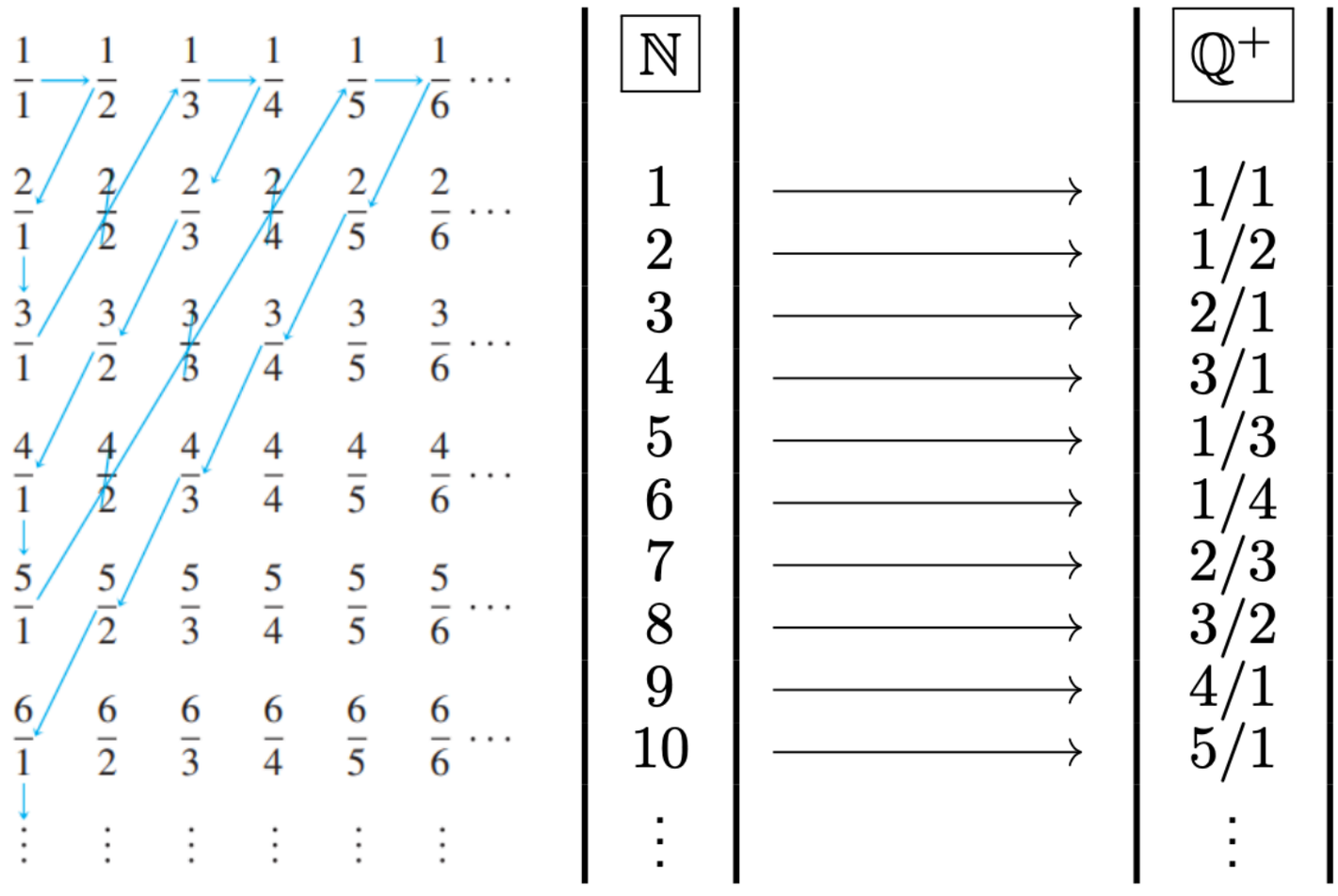
- Prove that the set of positive rational numbers are countable.

Set of positive rationals is countable

Problem

- Prove that the set of positive rational numbers are countable.

Solution



Set of positive rational numbers is countable

Problem

- Prove that the set of positive rational numbers are countable.

Solution (continued)

- To prove that $|\mathbb{N}| = |\mathbb{Q}^+|$, we need to prove that there is a one-to-one correspondence, say f, between $\mathbb{N}$ and $\mathbb{Q}^+$.
- Prove that f is onto.
  Every positive rational number appears somewhere in the grid.
  Every point in the grid is reached eventually.
- Prove that f is one-to-one.
  Skipping numbers that have already been counted ensures that no number is counted twice.

Example 4

Set of real numbers in $[0, 1]$ is uncountable

Problem

- Prove that the set of all real numbers between 0 and 1 is uncountable.

Set of real numbers in $[0, 1]$ is uncountable

Problem

- Prove that the set of all real numbers between 0 and 1 is uncountable.

Solution

- To prove that $|\mathbb{N}| \neq |[0..1]|$, we need to prove that there is no one-to-one correspondence between $\mathbb{N}$ and $[0..1]$.
- A powerful approach to prove the theorem is: proof by contradiction.

Set of real numbers in $[0, 1]$ is uncountable

Problem

- Prove that the set of all real numbers between 0 and 1 is uncountable.

Solution

Proof by contradiction.

- Suppose $[0..1]$ is countable.
- We will derive a contradiction by showing that there is a number in $[0..1]$ that does not appear on this list.

$\boxed{\mathbb{N}}$		$\boxed{[0..1]}$
1	$\longrightarrow$	$0.a_{11}a_{12}a_{13}\ldots a_{1n}\ldots$
2	$\longrightarrow$	$0.a_{21}a_{22}a_{23}\ldots a_{2n}\ldots$
3	$\longrightarrow$	$0.a_{31}a_{32}a_{33}\ldots a_{3n}\ldots$
$\vdots$	$\vdots$	$\vdots$
n	$\longrightarrow$	$0.a_{n1}a_{n2}a_{n3}\ldots a_{nn}\ldots$
$\vdots$	$\vdots$	$\vdots$

Set of real numbers in $[0, 1]$ is uncountable

Solution (continued)

- Suppose the list of reals starts out as follows:

0.	**9**	0	1	4	8	...
0.	1	**1**	6	6	6	...
0.	0	3	**3**	5	3	...
0.	9	6	7	**2**	6	...
0.	0	0	0	3	**1**	...
⋮	⋮	⋮	⋮	⋮	⋮	⋱

- Construct a new number $d = 0.d_1d_2d_3 \dots d_n \dots$ as follows:

$$d_n = \begin{cases} 1 & a_{nn} \neq 1, \\ 2 & a_{nn} = 1. \end{cases}$$

- We have $d = 0.12112\dots$, i.e.,

0.	**1**	**2**	**1**	**1**	**2**	...

Set of real numbers in $[0, 1]$ is uncountable

Solution (continued)

- Observation:
  For each natural number n, the constructed real number d differs in the nth decimal position from the nth number on the list.

1	$\longrightarrow$	0.	**9**	0	1	4	8	$\ldots$
2	$\longrightarrow$	0.	1	**1**	6	6	6	$\ldots$
3	$\longrightarrow$	0.	0	3	**3**	5	3	$\ldots$
4	$\longrightarrow$	0.	9	6	7	**2**	6	$\ldots$
5	$\longrightarrow$	0.	0	0	0	3	**1**	$\ldots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$
d	$\longrightarrow$	0.	**1**	**2**	**1**	**1**	**2**	$\ldots$

- This implies that d is not on the list. But, $d \in [0, 1]$.
- Contradiction! So, our supposition is false.
- Set of real numbers in $[0, 1]$ is uncountable.

There are different types of ∞!



Accepting a lemma in this course

- Given a and b, the open and closed intervals [a, b], (a, b), [a, b) and (a, b] all have the same cardinality

- Details: https://planetmath.org/openandclosedintervalshavethesamecardinality

Exercises

Exercise 1 (20 points)

The *sigmoid* function plays a pivotal role in machine learning. Particularly, it's instrumental in classification problems where we map predicted values to probabilities. The sigmoid function can squish any real-valued number into the range between 0 and 1, making it extremely useful for converting values into probabilities.

The sigmoid function S: $\mathbb{R} \to (0, 1)$, is defined as:

$$S(x) = \frac{1}{1 + e^{-x}}$$

Your task in this exercise is to check if this sigmoid function S is bijective. In other words, determine whether it's both injective (or one-to-one) and surjective (or onto).

As a reminder:

- Injectivity: Show that if S(x) = S(y), then x = y. This means that no two different inputs will yield the same output.
- Surjectivity: Show that for any number y in the range (0, 1), there is an x in the domain of real numbers such that S(x) = y. This means that every possible output is produced by some input.

- Prove real numbers and positive real number are of the same cardinality

- Find a bijection between [1,2] and [1,4] to show they are of the same cardinality

3. Let $3\mathbf{Z} = \{n \in \mathbf{Z} \mid n = 3k, \text{ for some integer } k\}$. Prove that $\mathbf{Z}$ and $3\mathbf{Z}$ have the same cardinality.

4. Let $\mathbf{O}$ be the set of all odd integers. Prove that $\mathbf{O}$ has the same cardinality as $2\mathbf{Z}$, the set of all even integers.

True or false, and proof

(h) [1 point] The size of the set of all multiples of 6 is less than the size of the set of all multiples of 3.

(i) [1 point] The size of the set of rational numbers is the same as the size of the set of real numbers in the range $[0, 0.0000001]$.

(j) [1 point] The size of the set of real numbers in the range $[1, 2]$ is the same or larger than the size of the set of real numbers in the range $[1, 4]$.